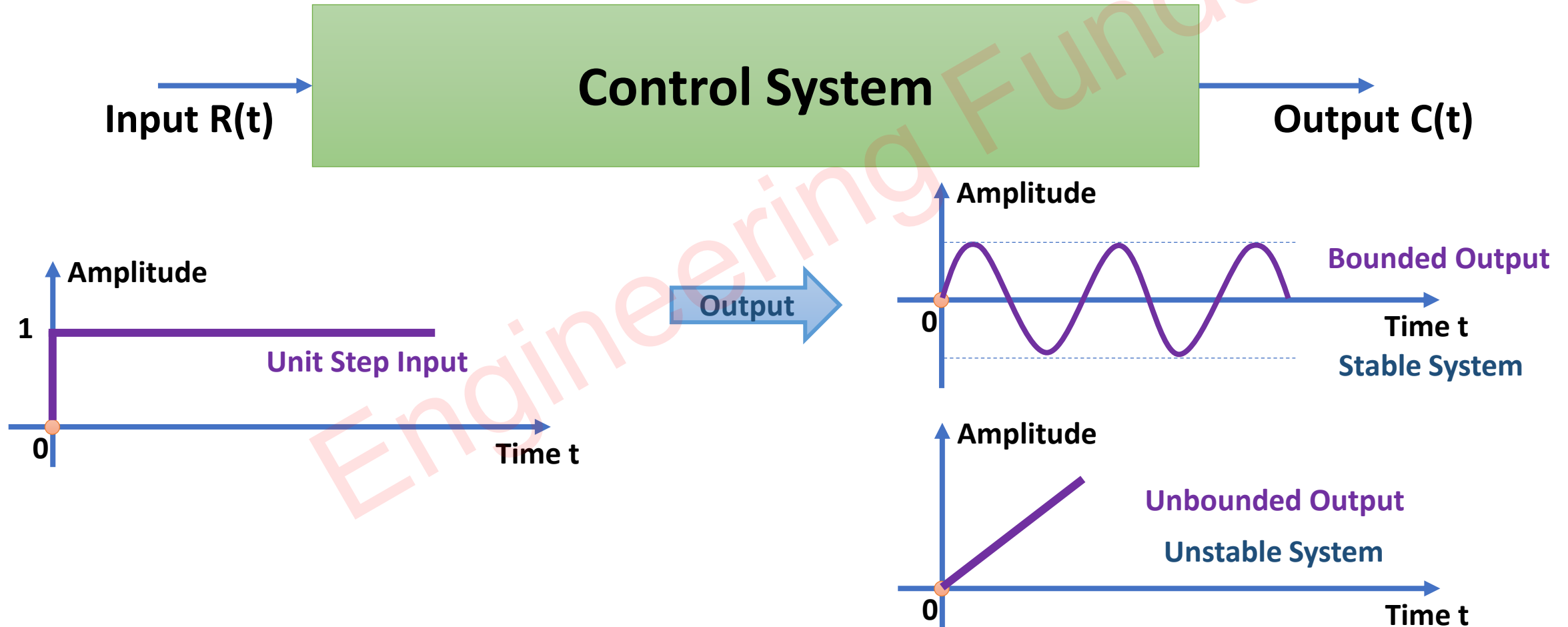


Stability Criteria for System

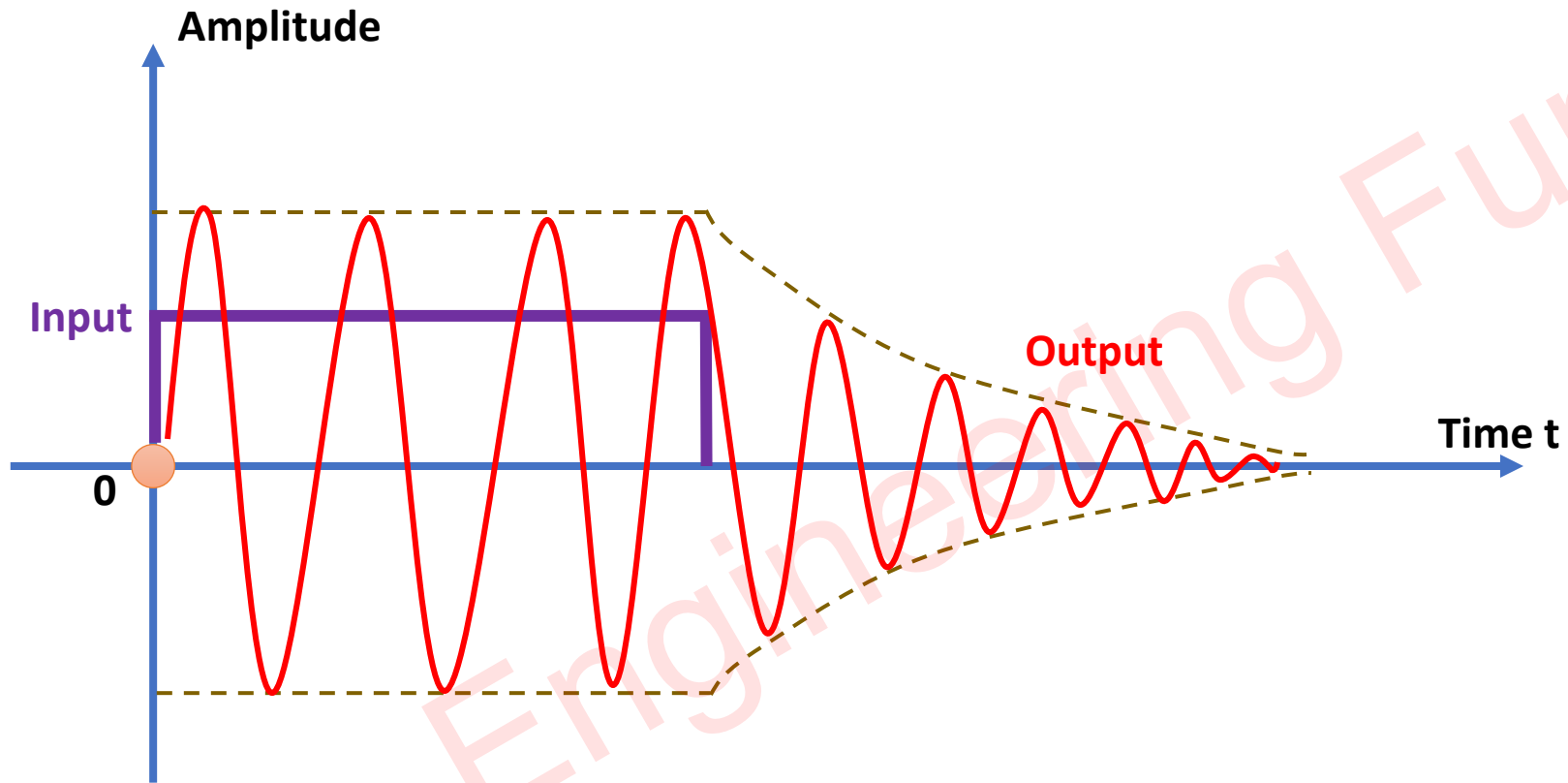
1st Stability Criteria for System

- A system is stable if the output is bounded with respect to the bounded input.



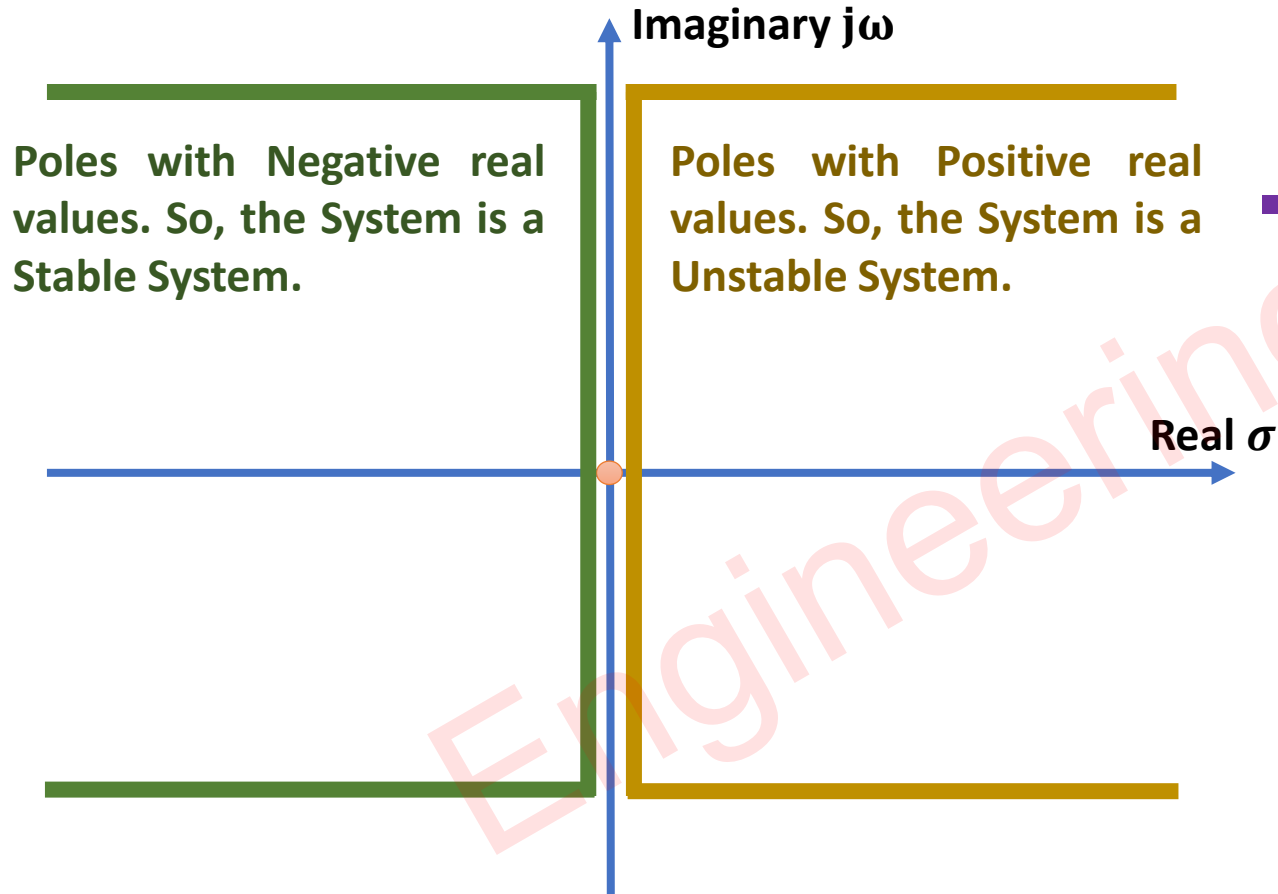
2nd Stability Criteria for System

- A system is Asymptotically stable if The output tends towards zero in the absence of Input. Irrespective of initial conditions.



3rd Stability Criteria for System

- The stability of the System depends on Poles. If all the poles are located in the left half of the S-Plane, then the system is a Stable System.



Poles with Negative real values. So, the System is a Stable System.

Poles with Positive real values. So, the System is a Unstable System.

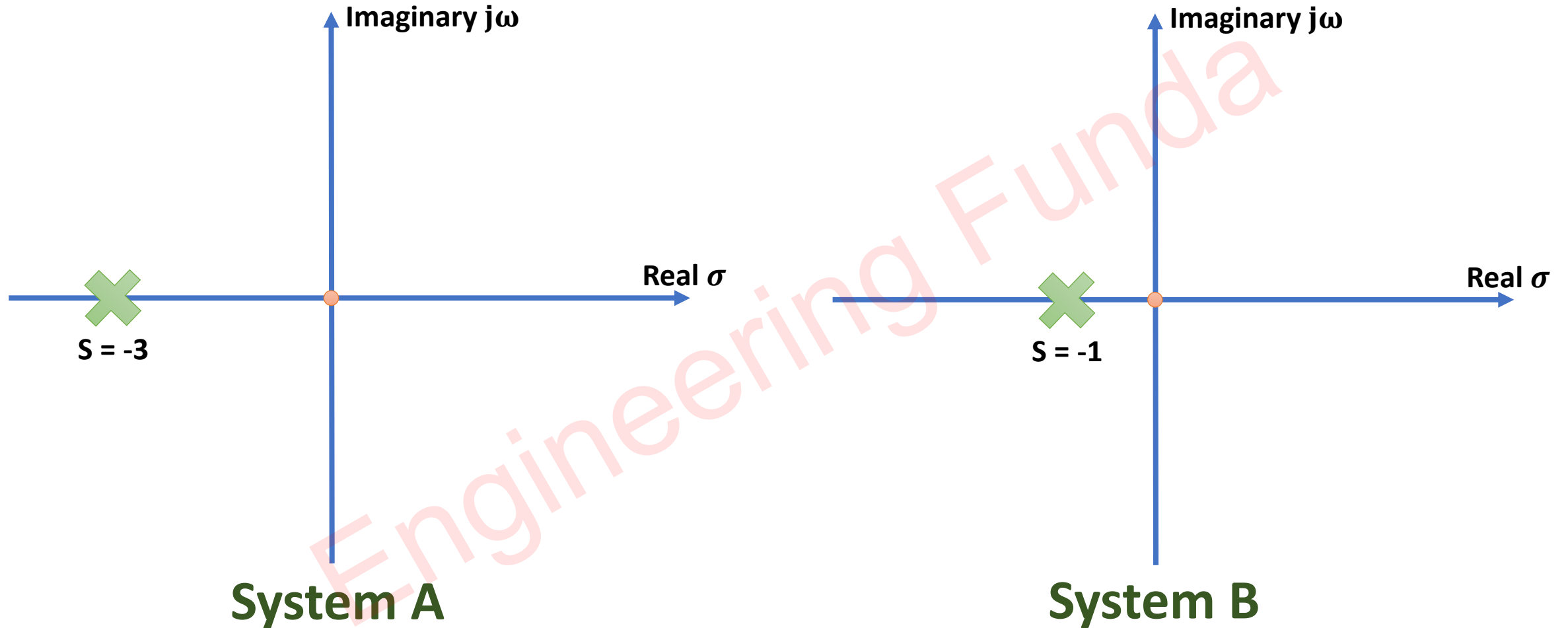
For Poles with Zero real values, System is Marginally Stable.

$$\Rightarrow T(s) = \frac{C(S)}{R(S)} = \frac{N(S)}{D(S)}$$

- $N(S)$ roots gives Zeros and $D(S)$ roots gives Poles.

4th Stability Criteria for System

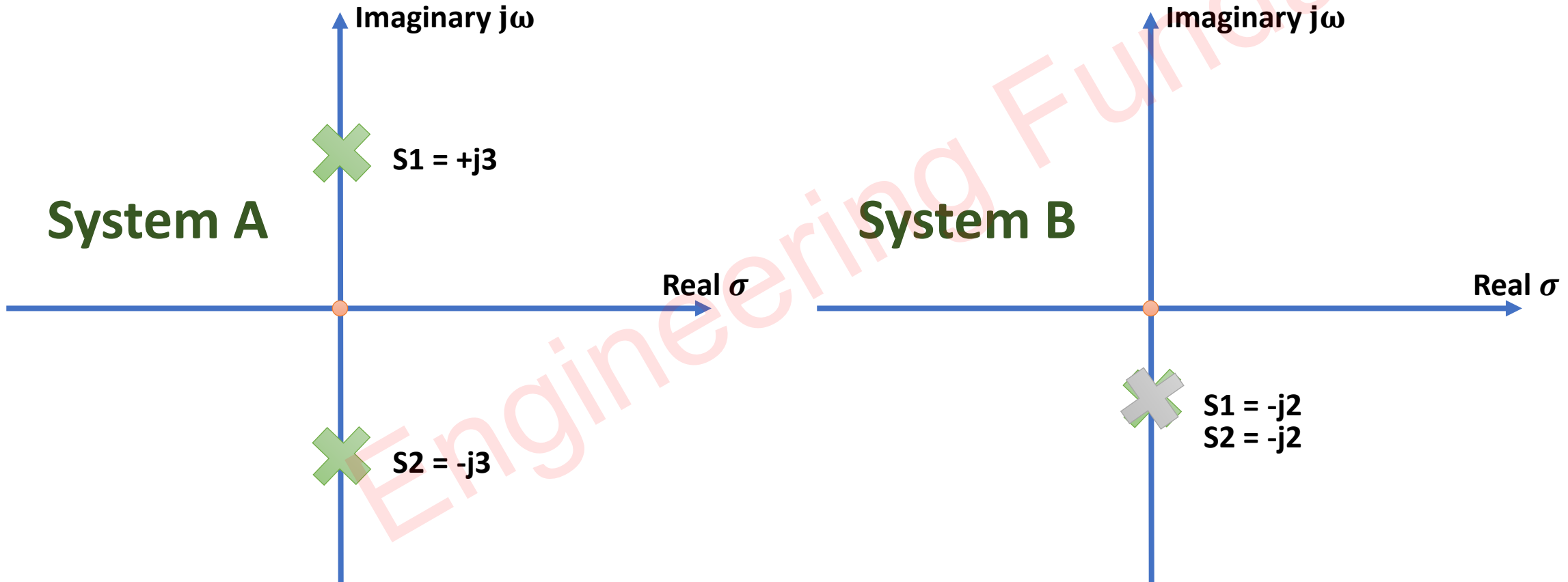
- As Poles approaches zero, The Stability Decreases.



- System A has higher stability compared to System B.

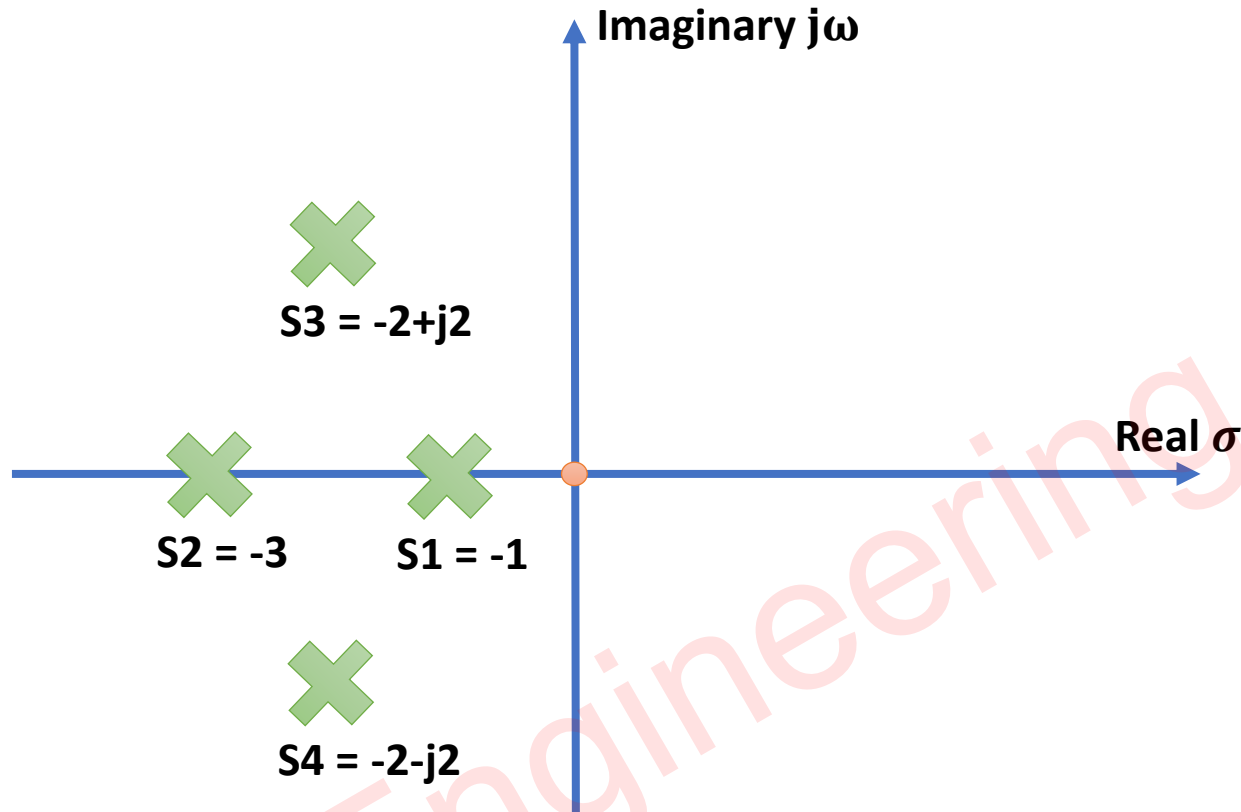
5th Stability Criteria for System

- When Poles are located on the Imaginary Axis. Then the system is a Marginally Stable System.
- These Poles should not be repeated on Imaginary Axis for Stability.



6th Stability Criteria for System

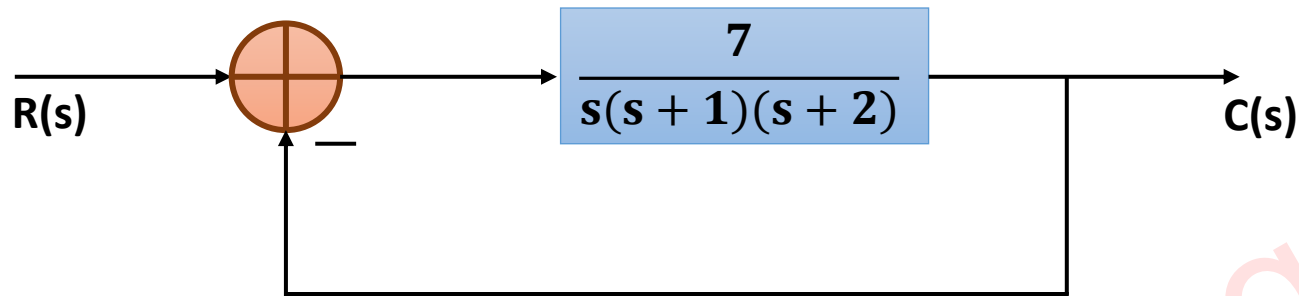
- The Poles which are close to origin are called dominant poles.



- Here, the pole at $S1 = -1$ is a dominant pole, as it most affects stability.
- **Robust stability** in a control system is the ability of a system to remain stable even when there are variations in its parameters or uncertainties. This ensures that the system operates reliably under different conditions.

Routh Stability Criteria

Question 1 — Check whether the given system is stable or unstable by Routh Stability Criteria.



∴ characteristics eqⁿ

$$\Rightarrow 1 + G(s)H(s) = 0$$

$$\Rightarrow 1 + \frac{7}{s(s+1)(s+2)} = 0$$

$$\Rightarrow s(s+1)(s+2) + 7 = 0$$

$$\Rightarrow s^3 + 3s^2 + 2s + 7 = 0$$

Routh matrix

s^3	1	2
s^2	3	7
s^1	$-\frac{1}{3}$	0
s^0	7	

$$= \frac{3 \times 2 - 1 \times 7}{3} = -\frac{1}{3}$$

$$= \frac{\left(-\frac{1}{3}\right) 7 - 3(0)}{-\frac{1}{3}} = 7$$

Check 1st column

- Two sign change $\rightarrow$ Two roots in RHP $\rightarrow$ Unstable

Question 2 — Using the Routh Stability Criterion, Determine the stability of the system described by characteristics equation:

$$\Rightarrow S^4 + 2S^3 + 8S^2 + 4S + 3 = 0$$

Routh matrix

S^4	1	8	3
S^3	2	4	
S^2	6	3	
S^1	3		
S^0	3		

- Check 1st Column

- No sign change $\rightarrow$ No root in RHP $\rightarrow$ Stable

$$= \frac{2 \times 3 - 1(4)}{2} = 3$$

$$= \frac{2 \times 8 - 1(4)}{2} = 6$$

$$= \frac{6(4) - 2(3)}{6} = 3$$

Question 3 — Check the stability of the system with the Routh Criterion for the characteristics equation given as follows:

$$\Rightarrow s^4 + s^3 + 20s^2 + 9s + 100 = 0$$

Routh matrix

s^4	1	20	100
s^3	1	9	
s^2	11	100	
s^1	$-\frac{1}{11}$		
s^0	100		

Check 1st Column

- Two time sign change $\rightarrow$ Two roots in RHP $\rightarrow$ Unstable

$$= \frac{1(20) - 1(9)}{1} = 11$$
$$= \frac{11(9) - 1(100)}{11} = -\frac{1}{11}$$

Routh Stability Criteria Special Case I

Question 1 — By Routh Criterion, Determine the Stability of the system described by the characteristics equation:

$$\Rightarrow s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$$

Routh matrix

s^5	1	2	3
s^4	1	2	5
s^3	$+ \boxed{0} \varepsilon$	-2	
s^2	$+ \boxed{\infty} \frac{2\varepsilon + 2}{\varepsilon}$	5	
s^1	$\oplus \frac{-5\varepsilon^2 - 4\varepsilon - 4}{2\varepsilon + 2}$	0	
s^0	5		

$$\frac{1(2) - 2(1)}{1} = 0$$

$$\frac{1(3) - 1(5)}{1} = -2$$

$$\frac{0(2) - (-2)(1)}{0} = \infty$$

$$\frac{\left(\frac{2\varepsilon + 2}{\varepsilon}\right)(-2) - 5\varepsilon}{\left(\frac{2\varepsilon + 2}{\varepsilon}\right)} = \frac{-5\varepsilon^2 - 4\varepsilon - 4}{2\varepsilon + 2}$$

$$\leadsto \lim_{\varepsilon \rightarrow 0} \frac{2\varepsilon + 2}{\varepsilon} = \frac{0 + 2}{0} = +\infty$$

$$\leadsto \lim_{\varepsilon \rightarrow 0} \frac{-5\varepsilon^2 - 4\varepsilon - 4}{2\varepsilon + 2} = \frac{-4}{2} = -2$$

$\leadsto$ In 1st column, two sign change = Two poles in RHP
 $\downarrow$
Unstable.

Question 2 — By Routh Criterion, Determine the Stability of the system described by the characteristics equation:

$$\Rightarrow s^5 + 2s^4 + 3s^3 + 6s^2 + 2s + 1 = 0$$

Routh Matrix

s^5	1	3	2
s^4	2	6	1
s^3	$+ \boxed{0} \varepsilon$	1.5	
s^2	$\ominus \boxed{\infty} \frac{6\varepsilon - 3}{\varepsilon}$	1	
s^1	$\oplus - \frac{\varepsilon^2 + 9\varepsilon - 4.5}{6\varepsilon - 3}$		
s^0	$\oplus 1$		

Arrows indicate the calculation of each row from the two rows immediately above it. For example, the s^3 row is calculated from s^5 and s^4 using the formula $\frac{2 \times 3 - 1 \times 6}{2} = 0$.

$$= \frac{2 \times 3 - 1 \times 6}{2} = 0$$

$$= \frac{2(2) - 1(1)}{2} = 1.5$$

$$= \frac{6(6) - 2(1.5)}{0} = \infty$$

$$= \frac{6\varepsilon - 2(1.5)}{\varepsilon}$$

$$= \frac{\left(\frac{6\varepsilon - 3}{\varepsilon} \right) 1.5 - \varepsilon}{\left(\frac{1\varepsilon - 3}{\varepsilon} \right)} = \frac{-\varepsilon^2 + 9\varepsilon - 4.5}{6\varepsilon - 3}$$

$$- \lim_{\varepsilon \rightarrow 0} \frac{6\varepsilon - 3}{\varepsilon} = \frac{0 - 3}{0} = -\infty$$

$$- \lim_{\varepsilon \rightarrow 0} \left(\frac{-\varepsilon^2 + 9\varepsilon - 4.5}{6\varepsilon - 3} \right) = \frac{-4.5}{-3} = +1.5$$

* In 1st column, Two sign change = Two poles in R.H.P.

↓
Unstable system.

Routh Stability Criteria Special Case II

Question 1 — By Routh Criterion, Determine the Stability of the system described by the characteristics equation:

$$\Rightarrow S^6 + 2S^5 + 8S^4 + 12S^3 + 20S^2 + 16S + 16 = 0$$

Routh matrix

S^6	1	8	20	16
S^5	2	12	16	
S^4	2	12	16	
S^3	8	24		
S^2	0	0	0	
S^1	6	16		
S^0	8/3	0		
	16			

$$\frac{(2)8 - 1(12)}{2} = 2$$

$$\frac{(2)20 - 1(16)}{2} = 12$$

$$\frac{8(12) - 2(24)}{8} = 6$$

$$\frac{6(24) - 8(16)}{6} = \frac{8}{3}$$

→ Auxillary eqⁿ: $A(s)$

$$\Rightarrow A(s) = 2s^4 + 12s^2 + 16$$

→ Differentiate $A(s)$

$$\Rightarrow \frac{dA(s)}{ds} = 8s^3 + 24s$$

→ Check 1st column,

- No sign change → Zero pole on RHP → Stable system.

Question 2 — By Routh Criterion, Determine the Stability of the system described by the characteristics equation:

$$\Rightarrow 2s^5 + s^4 + 4s^3 + 2s^2 + 2s + 1 = 0$$

Routh matrix

s^5	2	4	2
s^4	1	2	1
s^3	4	4	
s^2	0	0	0
s^1	1	1	
s^0	1		

Handwritten annotations: Arrows show the construction of the Routh matrix. The s^3 row is crossed out. The s^2 row is boxed. The s^1 row is boxed. The s^0 row is boxed.

$$\frac{1(4) - 2(2)}{1} = 0$$

$$\frac{1(2) - 2(1)}{1} = 0$$

$$\frac{4(2) - 1(4)}{4} = 1$$

$$\frac{4(1) - 4(1)}{1} = 0$$

~ Auxillary eq.ⁿ $A_1(s)$

$$\Rightarrow A_1(s) = s^4 + 2s^2 + 1$$

~ Differentiate $A_1(s)$

$$\Rightarrow \frac{dA_1(s)}{ds} = 4s^3 + 4s$$

~ Check 1st column

- Zero sign change $\rightarrow$ Zero poles on RHP $\rightarrow$ stable system.

~ Auxillary eq.ⁿ $A_2(s)$

$$\Rightarrow A_2(s) = s^2 + 1$$

~ Differentiate $A_2(s)$

$$\Rightarrow \frac{dA_2(s)}{ds} = 2s$$

Routh Stability Criteria {Stability Range}

Question 1 — Determine the Range of K to have a stable system by Routh Stability Criteria.

$$\Rightarrow s^3 + 8s^2 + 2s + 4K = 0$$

Routh matrix

$$\begin{array}{c|cc} s^3 & + & 1 \\ s^2 & + & 8 \\ s^1 & + & \frac{16-4K}{8} \\ s^0 & + & 4K \end{array}$$


$$= \frac{8(2) - 1(4K)}{8}$$

$$\boxed{0 < K < 4}$$

$$\Rightarrow \frac{16-4K}{8} > 0 \quad \Rightarrow 4K > 0$$

$$\Rightarrow \boxed{K > 0}$$

$$\Rightarrow 16 > 4K$$

$$\Rightarrow \boxed{4 > K}$$

Question 2 — If the System is given by

$$\Rightarrow G(S)H(S) = \frac{K}{(S+1)(S+3)(S+5)(S+7)}$$

Find the Range of K for a Stable System.

~> Characteristic eq.ⁿ

$$\Rightarrow 1 + G(S)H(S) = 0$$

$$\Rightarrow 1 + \frac{K}{(S+1)(S+3)(S+5)(S+7)} = 0$$

$$\Rightarrow (S+1)(S+3)(S+5)(S+7) + K = 0$$

$$\Rightarrow S^4 + S^3(7 + S + 3 + 1) + S^2(3S + 21 + 1S + 7 + S + 3) + S(10S + 35 + 21 + 1S) + 105 + K = 0$$

$$\Rightarrow S^4 + 16S^3 + 86S^2 + 176S + 105 + K = 0$$

Routh matrix

s^4	+	1	$\swarrow \searrow$	86
s^3	+	16	$\swarrow \searrow$	176
s^2	+	75	$\swarrow \searrow$	$105 + K$
s^1	+	$153.6 - 0.2133K$		
s^0	+	$105 + K$		

$$105 + K$$

$$= \frac{16(86) - 1(176)}{16} = 75$$

$$= \frac{75(176) - 16(105 + K)}{75}$$

$$= 153.6 - 0.2133K$$

$$\rightarrow 153.6 - 0.2133K > 0$$

$$\Rightarrow 153.6 > 0.2133K$$

$$\Rightarrow 720 > K$$

$$\rightarrow 105 + K > 0$$

$$\Rightarrow 105 > -K$$

$$\Rightarrow K > -105$$

$$\boxed{-105 < K < 720}$$

Routh Stability Criteria {Stability Optimization}

Question 1 — If the system is given by characteristics equation:

$$\Rightarrow S^3 + 5S^2 + 25S + 30 = 0$$

1. Find the Stability of the System.
2. Also Find Stability, If the roots change by $Z - 1$.
3. Also Find Stability, If the roots change by $Z - 2$.

Routh matrix

$$= \frac{5(25) - 1(30)}{5} = 19$$

S^3	+	1		25
S^2	+	5		30
S^1	+	19		
S^0	+	30		

(Note: In the original image, arrows indicate the calculation of the S^1 row element: $1 \times 30 - 5 \times 25 = -119$, but the handwritten value is 19. The diagram shows a cross between the 1 and 30, and the 5 and 25.)

∴ In 1st column, No sign change → all roots in LHP → Stable System.

→ For $z=1$, characteristic eq.ⁿ

$$\Rightarrow (z-1)^3 + 5(z-1)^2 + 25(z-1) + 30 = 0$$

$$\Rightarrow \underline{z^3} - \underline{1} - \underline{3z^2} + \underline{3z} + \underline{5z^2} - \underline{10z} + \underline{5} + \underline{25z} - \underline{25} + \underline{30} = 0$$

$$\Rightarrow z^3 + 2z^2 + 18z + 9 = 0$$

Routh matrix

z^3	+	1	↖	18
z^2	+	2	↗	9
z^1	+	13.5		
z^0	+	9		

$$\frac{2(18) - 1(9)}{2} = 13.5$$

→ 1st Column, No sign change → all roots in LHP → Stable system.

∴ For 2-2, characteristic eq.ⁿ

$$\Rightarrow (z-2)^3 + 5(z-2)^2 + 25(z-2) + 30 = 0$$

$$\Rightarrow \underline{z^3} - \underline{8} - \underline{6z^2} + \underline{12z} + \underline{5z^2} - \underline{20z} + \underline{20} + \underline{25z} - \underline{50} + \underline{30} = 0$$

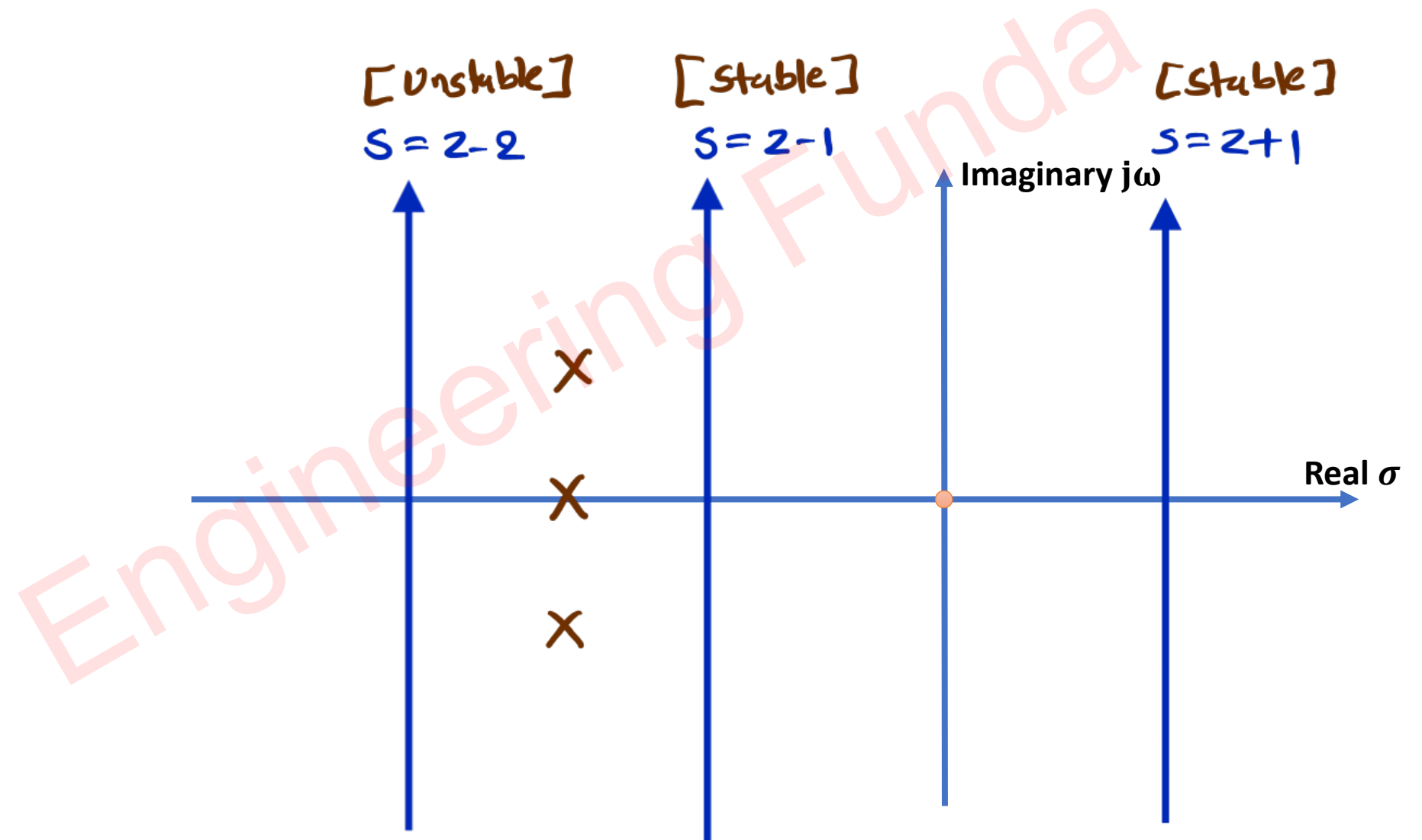
$$\Rightarrow z^3 - z^2 + 17z - 8 = 0$$

Routh matrix

	z^3	1	17
	z^2	-1	-8
	z^1	9	
	z^0	-8	

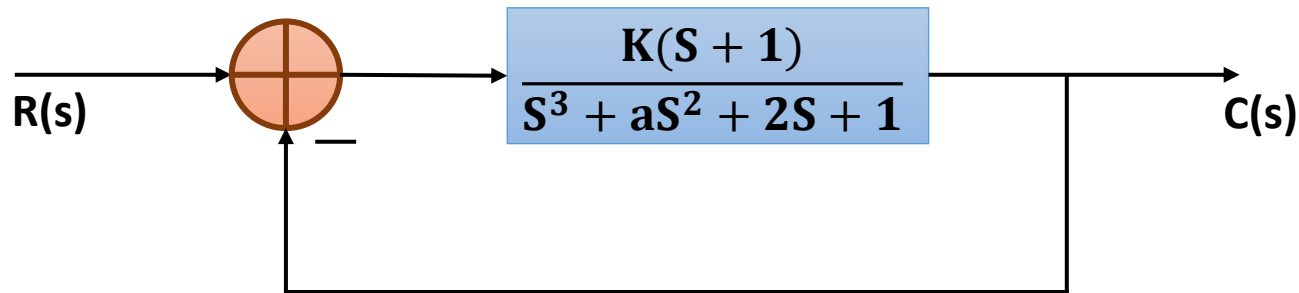
$$= \frac{(-1)17 - 1(-8)}{-1} = 9$$

∴ In 1st column, 3 sign change = 3 Poles in RHP → Unstable System



Routh Stability Criteria {Oscillatory Condition}

Question 1 — If the system given below oscillates at 2 rad/sec when $K = ?$ and $a = ?$



$= 2$

$= 0.75$

∴ Characteristics eq.ⁿ

$$\Rightarrow 1 + G(s)H(s) = 0$$

$$\Rightarrow 1 + \frac{K(S+1)}{S^3 + aS^2 + 2S + 1} = 0$$

$$\Rightarrow S^3 + aS^2 + 2S + 1 + K(S+1) = 0$$

$$\Rightarrow S^3 + aS^2 + S(2+K) + (1+K) = 0$$

Routh matrix

$$\begin{array}{c|cc} s^3 & + & 1 & 2+k \\ s^2 & + & a & 1+k \\ s^1 & + & \frac{a(2+k) - (1+k)}{a} & 0 \\ s^0 & + & 1+k & \end{array}$$

$$- \boxed{a > 0}$$

$$- \frac{a(2+k) - (1+k)}{a} > 0$$

$$\Rightarrow a(2+k) > (1+k)$$

$$\Rightarrow \boxed{a > \left(\frac{1+k}{2+k} \right)}$$

$$- 1+k > 0$$

$$\Rightarrow \boxed{k > -1}$$

~) For Oscillation take s^2 term.

$$\Rightarrow as^2 + (1+k) = 0$$

$$\Rightarrow \left(\frac{1+k}{2+k} \right) s^2 + (1+k) = 0$$

$$\Rightarrow s^2 = -(2+k)$$

$$\Rightarrow s^2 = j^2(2+k)$$

$$\Rightarrow s = \pm j\sqrt{2+k}$$

$$\Rightarrow \sigma \pm j\omega = \pm j\sqrt{2+k}$$

$$\Rightarrow \boxed{\sigma = 0} \quad \& \quad \omega = \sqrt{2+k}$$

$$\Rightarrow 2 = \sqrt{2+k}$$

$$\Rightarrow 4 = 2+k$$

$$\Rightarrow \boxed{k = 2}$$

$$\Rightarrow a > \frac{1+k}{2+k}$$

$$\Rightarrow a > \frac{1+2}{2+2}$$

$$\Rightarrow \boxed{a > 0.75}$$

Routh Stability Criteria

Question 1 — Find The number of roots in the Left Half Plane for the given characteristics equation:

$$\Rightarrow s^3 + 5s^2 + 7s + 3 = 0$$

Routh matrix

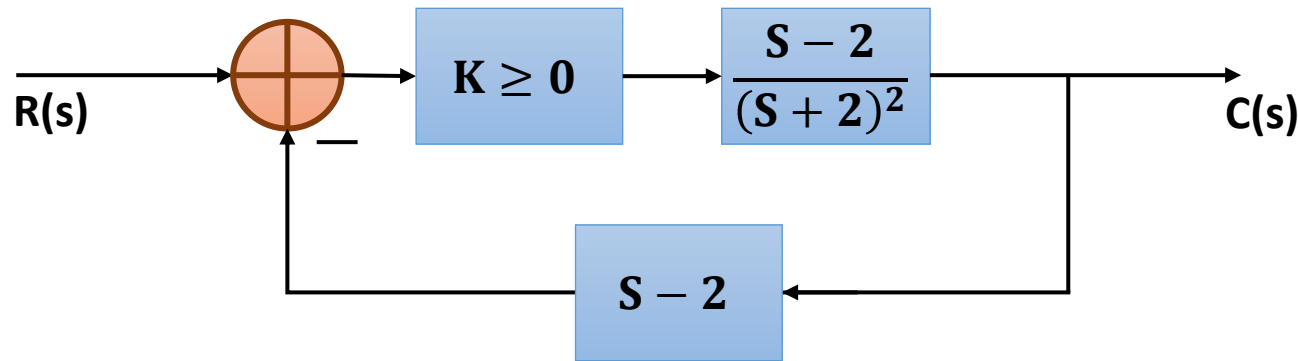
s^3	+	1	$\swarrow$	$\searrow$	7
s^2	+	5	$\swarrow$	$\searrow$	3
s^1	+	$32/5$			0
s^0	+	3			

$$= \frac{5(7) - 1(3)}{5} = \frac{32}{5}$$

$\Rightarrow$ In 1st column, No sign change $\rightarrow$ zero poles in RHP.

$$\begin{aligned} \rightarrow \text{Poles in LHP} &= \text{Total poles} - \text{Poles in RHP} \\ &= 3 - 0 = 3 \end{aligned}$$

Question 2 — Find the Range of K for a Stable System.



$$- G(s) = \frac{k(s-2)}{(s+2)^2}$$

$$- H(s) = s-2$$

⇒ Characteristics eqⁿ

$$\Rightarrow 1 + G(s)H(s) = 0$$

$$\Rightarrow 1 + \left(\frac{k(s-2)}{(s+2)^2} \right) (s-2) = 0$$

$$\Rightarrow (s+2)^2 + k(s-2)^2 = 0$$

$$\Rightarrow s^2 + 4s + 4 + ks^2 - 4ks + 4k = 0$$

$$\Rightarrow s^2(1+k) + s(4-4k) + (4+4k) = 0$$

Routh matrix

$$\begin{array}{c|c} s^2 & 1+k \\ s^1 & 4-4k \\ s^0 & 4+4k \end{array}$$

$$4+4k$$

$$0$$

Range of k

$$- 1+k > 0 \Rightarrow \boxed{k > -1}$$

$$\begin{aligned} - 4-4k > 0 &\Rightarrow 4 > 4k \\ &\Rightarrow \boxed{1 > k} \checkmark \end{aligned}$$

$$\begin{aligned} - 4+4k > 0 &\Rightarrow 4k > -4 \\ &\Rightarrow \boxed{k > -1} \end{aligned}$$

$$\boxed{1 > k \geq 0}$$

Question 3 — An Electro-Mechanical Closed Loop Control System has the following characteristics equation:

$$\Rightarrow S^3 + 6KS^2 + (K + 2)S + 8 = 0$$

Where K is the forward gain of the System, find the value of K for a Stable System.

Routh matrix

$+ S^3$	1	$K+2$
$+ S^2$	$6K$	8
$+ S^1$	$\frac{3K^2+6K-4}{3K}$	0
$+ S^0$	8	

(Note: In the original image, arrows indicate the calculation of the third row from the first two rows.)

$$- 6K > 0$$

$$\Rightarrow \boxed{K > 0}$$

$$- \frac{3K^2+6K-4}{3K} > 0$$

$$\Rightarrow 3K^2+6K-4 > 0$$

$$\begin{aligned} &= \frac{6K(K+2) - 1(8)}{6K} = \frac{6K^2 + 12K - 8}{6K} \\ &= \frac{3K^2 + 6K - 4}{3K} \end{aligned}$$

$$- k = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-6 \pm \sqrt{36 + 4 \times 3 \times 4}}{2 \times 3}$$

$$= \frac{-6 \pm \sqrt{84}}{6}$$

$$= 0.527 \quad \text{or} \quad -2.53$$

$$- \boxed{k > 0.527} \quad \text{or} \quad \boxed{k > -2.53}^x$$

$$- \text{Range of } k \Rightarrow \boxed{k > 0.527}$$

Routh Stability Criteria

Question 4 — If the characteristics equation of the system is given by:

$$\Rightarrow S^3 + 3S^2 + 4S + A = 0$$

Then all the roots of this equation are in the left half of the S Plane providing that:

- a. $A > 12$
- b. $-3 < A < 4$
- ☒ c. $0 < A < 12$
- d. $5 < A < 12$

Routh matrix

S^3	+	1	↖	4
S^2	+	3	↗	A
S^1	+	$\frac{12-A}{3}$		0
S^0	+	A		

$$- A > 0$$

$$- \frac{12-A}{3} > 0 \Rightarrow 12 > A$$

$$\boxed{12 > A > 0}$$

Question 5 — If the characteristics equation of the system is given by:

$$\Rightarrow 15S^2 + 6S + K + 10 = 0$$

The Condition for stability is:

- a. $K > 5$
- ☒ b. $-10 < K$
- c. $K > -4$
- d. $-10 < K < -4$

Routh matrix

s^2	+	15	$K+10$
s^1	+	6	
s^0	+	$K+10$	

$$- K+10 > 0 \Rightarrow K > -10$$

Question 6 — If the characteristics equation of the system is given by:

$$\Rightarrow S^4 + 3S^3 + 5S^2 + 6S + K + 10 = 0$$

The Condition for stability is:

- a. $K > 5$
- b. $-10 < K$
- c. $K > -4$
- d. ✓ $-10 < K < -4$

Routh matrix

s^4	+	3	5
s^3	+	3	6
s^2	+	3	$K+10$
s^1	+	$-K-4$	0
s^0	+	$K+10$	

$$= \frac{3 \times 5 - 1(6)}{3} = 3$$

$$= \frac{3(6) - 3(K+10)}{3}$$

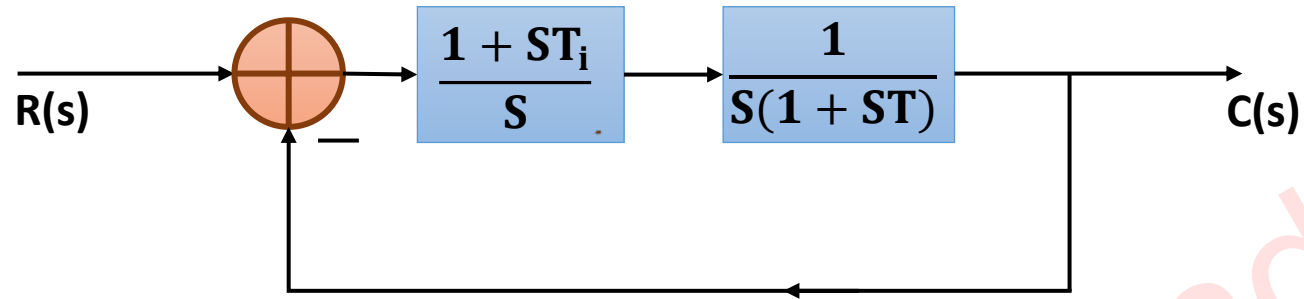
$$- (-K-4) > 0 \Rightarrow -4 > K$$

$$- K+10 > 0 \Rightarrow K > -10$$

$$\boxed{-4 > K > -10}$$

Question 7 — In Order to stabilize the system shown in the figure, T_i should satisfy

- a. $T_i = -T$
- b. $T_i = T$
- c. $T_i < T$
- d. ✓ $T_i > T$



⇒ Characteristic eqⁿ

$$\Rightarrow 1 + G(s)H(s) = 0$$

$$\Rightarrow 1 + \left(\frac{1 + sT_i}{s} \right) \left(\frac{1}{s(1 + sT)} \right) = 0$$

$$\Rightarrow s^2(1 + sT) + 1 + sT_i = 0$$

$$\Rightarrow s^2 + s^3T + 1 + sT_i = 0$$

$$\Rightarrow s^3T + s^2 + sT_i + 1 = 0$$

Routh matrix

s^3	+	T	T_i
s^2	+	1	1
s^1	+	$T_i - T$	0
s^0	+	1	



- $T > 0$

- $T_i - T > 0 \Rightarrow T_i > T$

Engineering Funda

Routh Stability Criteria

Question 8 — The Open Loop Transfer Function of an unity Negative Feedback system is given by:

$$\Rightarrow \frac{2s^2 + 6s + 5}{(s + 1)^2(s + 2)}$$

The characteristics equation of the closed-loop system is given by

Also, Check the Stability of the system.

↪ Characteristics eq.ⁿ

$$\Rightarrow 1 + G(s)H(s) = 0$$

$$\Rightarrow 1 + \left(\frac{2s^2 + 6s + 5}{(s + 1)^2(s + 2)} \right) = 0$$

$$\Rightarrow (s^2 + 2s + 1)(s + 2) + 2s^2 + 6s + 5 = 0$$

$$\Rightarrow s^3 + 2s^2 + 2s^2 + 4s + s + 2 + 2s^2 + 6s + 5 = 0$$

$$\Rightarrow s^3 + 6s^2 + 11s + 7 = 0$$

Routh matrix

s^3	+	1	↗ ↘	11
s^2	+	6	↗ ↘	7
s^1	+	$9/6$		0
s^0	+	7		

- In 1st column

↓

Zero sign change

↓

Stable system

↓

All roots in LHP (3 poles)

Question 9 — A described by the transfer function $H(S) = \frac{1}{S^3 + \alpha S^2 + KS + 3}$ is stable. The Conditions for α and K are

- a. $\alpha > 0, \alpha K < 3$
- b. $\alpha > 0, \alpha K > \underline{3}$
- c. $\alpha < 0, \alpha K > 0$
- d. $\alpha < 0, \alpha K < 0$

$\Rightarrow$ Characteristics eq.ⁿ

$$\Rightarrow S^3 + \alpha S^2 + KS + 3 = 0$$

Routh matrix

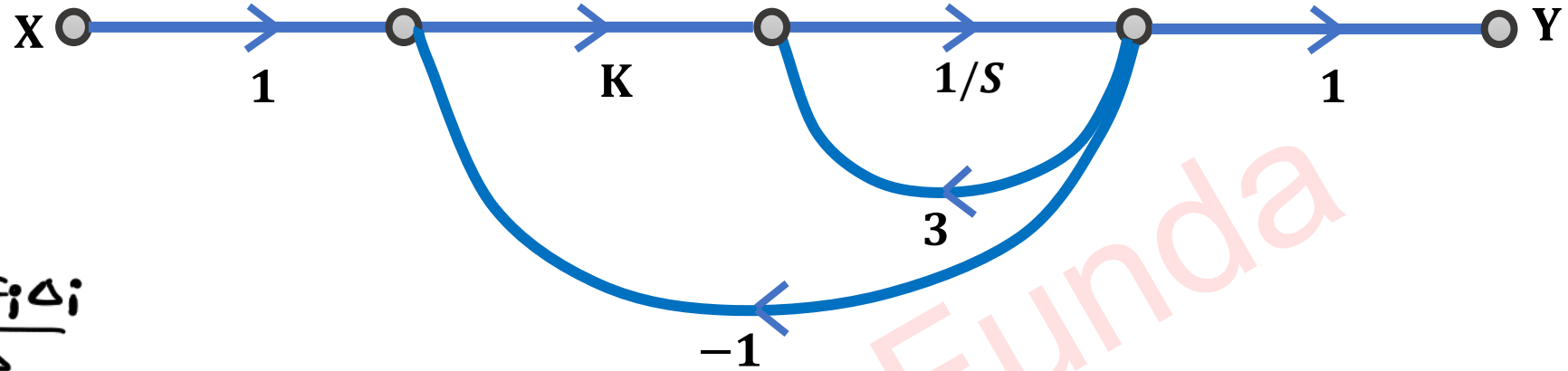
S^3		+	1	$\swarrow$	K
S^2		+	α	$\searrow$	3
S^1		+	$\frac{\alpha K - 3}{\alpha}$		0
S^0		+	3		

— $\alpha > 0$

— $\frac{\alpha K - 3}{\alpha} > 0 \Rightarrow \alpha K > 3$

Question 10 — The system shown in the figure remains stable when

- a. $K < -1$
- b. $-1 < K < 1$
- c. $1 < K < 3$
- d. $K > 3$



$$\Rightarrow \frac{Y}{X} = \frac{\sum F_i \Delta_i}{\Delta}$$

⇒ Forward paths

$$- F_1 = 1 \times K \times \frac{1}{s} \times 1 = \frac{K}{s}$$

→ Δ_i

$$- \Delta_1 = 1 - \sum \text{loops not touching } F_1 \\ = 1 - 0 = 1$$

⇒ Single loops

$$- L_1 = \frac{1}{s} \times 3 = \frac{3}{s}$$

→ Δ

$$- \Delta = 1 - \sum \text{single loops}$$

$$- L_2 = K \times \frac{1}{s} \times -1 = -\frac{K}{s}$$

$$= 1 - \left(\frac{3}{s} - \frac{K}{s} \right)$$

$$= 1 - \frac{3}{s} + \frac{K}{s}$$

Routh matrix

$$\begin{array}{c|c} + s^1 & 1 \\ + s^0 & -3+k \end{array}$$

$$- -3+k > 0$$

$$\boxed{\Rightarrow k > 3}$$

$$-\frac{Y}{X} = \frac{(K/s)}{1 - \frac{3}{s} + \frac{k}{s}} = \frac{k}{s-3+k}$$

$\leadsto$ characteristic eqⁿ

$$\Rightarrow s-3+k=0$$